

TEMASEK JUNIOR COLLEGE, SINGAPORE
JC 2
Preliminary Examinations 2013
Higher 1

MATHEMATICS

Paper 1

Additional Materials: Answer paper
List of Formulae (MF15)

8864/01

18 September 2013

3 hours

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.

Pure Mathematics [35 marks]

- 1 By using an algebraic method, solve the equation

$$3^{x+1} + 26 - 3^{2-x} = 0. \quad [4]$$

- 2 (i) Sketch the graphs of $y = (2x-1)(2x-9)$ and $y = \ln(2x-5)$ on the same diagram, giving the equations of any asymptotes, the coordinates of any turning points and the coordinates of any points of intersection with the x - and y -axes. [3]

- (ii) Hence solve the inequality $(2x-1)(2x-9) > \ln(2x-5)$ for $x > 3$. [2]

- 3 Alex wishes to construct a closed rectangular box of height y m with a horizontal square base of side x m. The material used to construct the top and bottom surfaces of the box costs \$0.50 per m^2 . The material used to construct the vertical surfaces of the box costs \$0.75 per m^2 .

- (i) Show that the total cost of constructing the box is given by C , where $\$C = x^2 + 3xy$. [1]

- (ii) It is required that y must exceed x by exactly 1 m. Given that Alex has a budget of \$100, determine the greatest value of x in exact form. [4]

- 4 An open cylindrical tank of internal radius r m and internal height h m is to be constructed such that it has an internal volume of 8 m^3 . Show that its internal surface area, $A \text{ m}^2$, is given by

$$A = \pi r^2 + \frac{16}{r}. \quad [2]$$

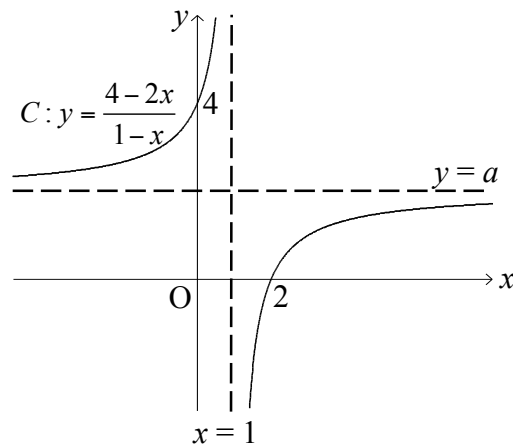
- (i) Find the radius of the tank for which the internal surface area is minimised, leaving your answer in exact form. [3]

- (ii) It is given that $r = 2$. When the tank is initially subjected to a heating process, its internal radius increases at a rate of $\frac{1}{10^6(\pi-1)}$ m per minute. Find the rate of increase of its internal surface area at this instant. [2]

- 5 (a) The gradient at any point (x, y) on a curve is given by $\sqrt{k-2x}$, where k is a constant. Given that the curve has a stationary point at $(2, 4)$, find the equation of the curve, giving the value of k . [4]

- (b) Find the integral $\int \frac{e^x - e^{2x/3}}{e^{3x}} dx$. [3]

6 Do not use a graphic calculator in answering this question.



The diagram shows the curve C with equation $y = \frac{4-2x}{1-x}$. The curve has asymptotes $x = 1$ and $y = a$.

- (i) Find the value of a . [1]
- (ii) Find the equation of the tangent to the curve at the point $(3, 1)$. [3]
- (iii) Find the exact value of the area bounded by C , its tangent at $(3, 1)$ and the x -axis. [3]

Statistics [60 marks]

- 7** An ice-cream manufacturer produces 4000 cups of ice-cream per day in 4 different flavours, with number of cups for each flavour as shown in the following table:

Flavour	Number of cups produced per day
Vanilla	1600
Chocolate	1200
Strawberry	800
Green Tea	400

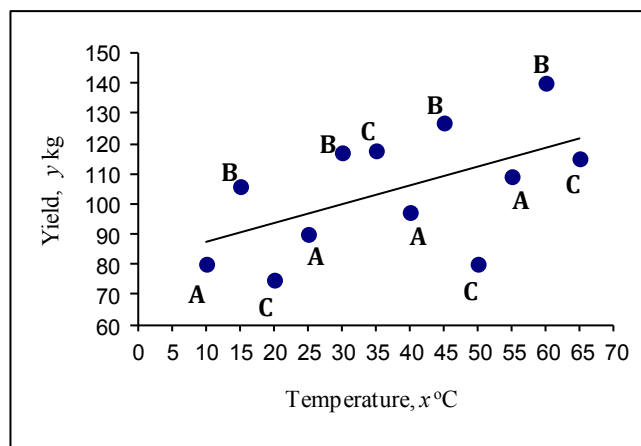
The production line operates from 9 am to 5 pm on weekdays. The quality control team conducts a daily check on the quality of ice-cream produced by taking a simple random sample of 50 cups of ice-cream for testing every day between 2 pm and 3 pm.

- (i) Give two reasons why this method of sampling is not satisfactory. [2]
- (ii) Suggest a better sampling method to select a sample of size 50 and describe how it can be carried out. [3]

- 8 Three trainee technicians carry out laboratory trials to examine the effect of temperature on the yield of an industrial process. The table below shows the results obtained by each technician.

Technician	A	B	C	A	B	C	A	B	C	A	B	C
Temperature, x °C	10	15	20	25	30	35	40	45	50	55	60	65
Yield, y kg	80	106	75	90	117	118	97	127	80	109	140	115

A scatter diagram of the above data, together with the least squares regression line of y on x , is shown below:



Using the above data,

- (i) calculate the equation of the regression line of y on x and give an interpretation of the regression coefficient, [3]
- (ii) find the product moment correlation coefficient between x and y . [1]

It is known that over this range of temperatures, the relationship between yield and temperature is approximately linear.

- (iii) Use the equation found in (i) to estimate the yield at a temperature of 52°C, giving your answer to the nearest kg. [1]
 Comment on the reliability of this estimate. [1]

An experienced and reliable technician carries out the trial at a temperature of 40°C and obtains a yield of 125 kg.

- (iv) Without further calculations, state which trainee's results are more consistent with that of the experienced technician. [1]
 Describe how you would improve the estimate for the yield at a temperature of 52°C without further data collection. [1]

- 9 A pharmaceutical company claims that the Vitamin C pills which it manufactures take an average of thirty minutes to dissolve individually.

- (a) Mr Grew believes that the claim is untrue. He measures the time taken, x min, for a hundred of these Vitamin C pills to dissolve individually, which yields the mean k min and variance 4.05 min^2 . A hypothesis test at the 10% level of significance indicates that Mr Grew's belief is valid. Determine the set of possible values of k . [6]

Explain, in the context of the question, the meaning of 'at the 10% level of significance'. [1]

- (b) Mr Minnion measures the time taken, x min, for eighty of these Vitamin C pills to dissolve individually, which yields the following results:

$$\sum (x - 30) = 128.6, \quad \sum (x - 30)^2 = 581.74.$$

Find unbiased estimates of the population mean and population variance of the time taken for a Vitamin C pill to dissolve individually. [3]

- 10 A confectionery Radulee bakes macarons. The mass of a lavender macaron follows a normal distribution with mean 12 g and standard deviation 0.2 g. The mass of a rose macaron follows a normal distribution with mean 11 g and standard deviation 0.1 g.

- (i) Find the probability that the difference in the masses of a lavender macaron and a rose macaron is at least 0.8 g. [4]

State an assumption that you have used in your calculations. [1]

- (ii) Find the probability that twice the mass of a lavender macaron exceeds the total mass of two rose macarons by more than 2.2 g. [3]

- (iii) Radulee discards lavender macarons with a mass of less than 11.5 g. Determine the median mass of a discarded lavender macaron, leaving your answer correct to four decimal places. [2]

- (iv) A customer orders two lavender macarons. Find the probability that one of them weighs more than 12 g while the other weighs less than 12 g. [2]

- 11** The height of Primary 1 pupils in a big town has a mean of 120 cm and a standard deviation of 10 cm. It is known that 45% of the pupils among the current cohort of Primary 1 pupils in the town are male.

A school in the town has an enrolment of 480 Primary 1 pupils.

- (i) Find the probability that the mean height of Primary 1 pupils in the school is more than 121 cm. Is it necessary to assume that the height of Primary 1 pupils has a normal distribution for your calculations to be valid? [3]
- (ii) Using a suitable approximation, find the probability that there are at most 250 Primary 1 female students in the school. [4]

The pupils in the school are allocated randomly to 12 classes of 40 each.

- (iii) Find the probability that there are more male students than female students in a class. [3]
- (iv) Find the probability that exactly two classes in the school have more male students than female students. [2]

- 12** In a large city, there was a surge in the sales of face masks and air purifiers due to a haze situation. A survey conducted by the Health Department showed that 27% of residents bought both masks and air purifiers. Among residents who bought face masks, 30% of them also bought air purifiers.

- (i) Find the probability that a randomly selected resident bought face masks. [2]
- (ii) Find the probability that out of three randomly selected residents, exactly two bought face masks. [2]

The survey also showed that among residents who did not buy face masks, 5% of them bought air purifiers.

- (iii) Show that the probability that a randomly selected resident bought air purifiers is 0.275. [3]
- (iv) Given that a randomly selected resident bought air purifiers, find the probability that he also bought face masks. [2]
- (v) Determine if the event that a randomly selected resident bought face masks and the event that a randomly selected resident bought air purifiers are
 - (a) mutually exclusive, [1]
 - (b) independent. [1]
- (vi) Find the probability that a randomly selected resident bought either face masks or air purifiers or both. [2]

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